

## Sheet 5 - decision theory

- a) Reflexivity: yes,  $x \leq x+1$  so  $x \succsim x \quad \forall x$   
 Connectedness: yes:  $\forall x \forall y \neq x$ , at least one of  $x \leq y+1$  or  $y \leq x+1$  holds, by ~~the~~ since each condition is a weakening of  $x \leq y$ ;  $y \leq x$ .  
**Q1a. yes, correct**  
 Completeness: yes, since reflexive and connected  
 Transitivity: no, e.g.  ~~$2 \succsim 1$  and  $3 \succsim 2$  but  $3 \not\succsim 1$~~   
 $2 \succsim 1$  and  $3 \succsim 2$  but  $3 \not\succsim 1$ .  
 Symmetry: no, e.g.  $5 \succsim 10$  and  $10 \not\succsim 5$   
 Antisymmetry: no, e.g.  $1 \succsim 2$  and  $2 \succsim 1$  with  $2 \neq 1$ .

- b) Not rational. Preferences  $\succsim$  are rational iff they satisfy completeness and transitivity, but here transitivity fails. **Q1b. yes, correct**  
 c) Not representable by utility function. If a preferences have such a representation,  $u$  s.t.  $x \succsim y \Leftrightarrow u(x) \geq u(y)$  then they must be rational, but these aren't. It's a contradiction to have such a  $u$  because we'd require  $u(2) \geq u(1)$ ,  $u(3) \geq u(2)$  but  $u(3) < u(1)$ . **Q1c. correct**

- 2a) Yes, by Debreu's theorem - since they're rational and continuous - they can be represented by a continuous  $u$ .  
 In particular, e.g.  $u(x_1, x_2) = x_1^3 x_2$  clearly works.

- b) Any strictly increasing transformation of a utility function represents the same preferences.  

$$v(x_1, x_2) = (u(x_1, x_2))^{1/4}$$
 which is strictly increasing,  $\checkmark$   

$$w(\cdot) = \log[u(\cdot)]$$
 likewise  $\checkmark$   

$$t(\cdot) = (u(\cdot))^{-1}$$
 which is strictly decreasing,  $\times$ .  
 since ~~exp.~~ The prefs are inverted exactly;  $-t(\cdot)$  would preserve them though!

- c) ~~We want some  $x$  s.t.  $(x_1, x_2) \sim (x_1^3, x) \Leftrightarrow x_1^3 x_2 = x_1^4$~~

a U s.t.

c) we want  $U(x_1, x_2) = x_1^x x_2^y$  where  $(x, x) \sim (x_1, x_2)$   
 $\Leftrightarrow x^4 = x_1^3 x_2$

So  $x = x_1^{3/4} x_2^{1/4}$ , i.e.  $U(x_1, x_2) = x_1^{3/4} x_2^{1/4}$ .

3. Mon:  $\{F, M, P\}$  Tue:  $\{P, V\}$  Wed:  $\{M, V\}$

a) Only Mon contains F so to pin it down in the ranking, we need it chosen that day (i.e. strictly best), and transitivity letting us compare it to V. So we need V to be dispreferred to at least one of P or M. And to pin down P vs M we need only exactly one of them chosen, so the non-chosen one is worst. i.e. possible choices are  $\frac{FVM}{(i)}$  or  $\frac{FPV}{(ii)}$

b) Directly revealed: (i)  $F \succ M, F \succ P, \{V \succ P \text{ and } M \succ V\}$  (ii)  $\{P \succ V + V \succ M\}$

Indirectly revealed:

(i)  $F \succ V, M \succ P$

(ii)  $F \succ V, P \succ M$

c) We'd know  $P \succ M, P \succ F, V \succ P, V \succ M$  directly; ~~and~~  $V \succ F$  indirectly. We want to compare  $\{M, F\}$  on Thu.

d)  $P \succ F, P \succ M, V \succ P, M \succ V$  but not rational! We have a cycle (or, perhaps prefs are only weakly/unstable)

e)  $M \rightarrow V \rightarrow P \rightarrow F$  ignoring M on Mon.

So they'd choose M on Thu.

4. a) A choice function is  $c: \Delta \rightarrow X$  where  $\Delta$  is <sup>some</sup> ~~the~~ set of possible menus (i.e.  $\Delta \subseteq 2^X \setminus \{\emptyset\}$ ) and  $X$  is the choice set, with  $c(B) \in B \quad \forall B \in \Delta$

In this case, we'd have  $X = \{A, B, C, D, 0\}$  and  $c(X) = 0$  while  $c(M) \in \{A, B, C, D\}$  for any menu  $M \neq X$ . (where menus must always include 0).

Or, if there are some shoes that the client actually prefers not to buy (vs 0) then the ~~preference~~ choice function would be

$$c(M) = \begin{cases} 0 & \text{if } |M| = 5 \\ \arg \max_{x \in M} u(x) & \text{otherwise.} \end{cases}$$

Henceforth I assume  $0 \leq 5$  for each shoe <sup>not needed</sup>

b)  $\alpha$ : if  $M \subseteq N$  and  $c(N) \in M$  then  $c(N) = c(M)$ .

But consider  $N = \{A, B, C, D, 0\}$  and e.g.  $M = \{A, B, C, 0\}$ .  
 $c(N) = 0$  but  $c(M) = \arg \max_{x \in M} u(x) \in \{A, B, C\} \neq 0$ .  
 So  $\alpha$  fails.

WARP: if  $x, y \in M$  and  $\in N$  then we can't have  $c(M) = x$  and  $c(N) = y$ .

But consider again the above menus  $M, N$ ;  
 let  $y = 0 = c(N) \neq c(M) = x := \arg \max_{x \in M} u(x)$ .  
 So WARP fails.

This behaviour is not rational; it's a failure of transitivity.

c) Showing four shoes leads to zero profit; certainly this is worse than showing only one, which at worst (if least-favourite) yields £5.

Showing two guarantees choice is at least third-<sup>favourite</sup> best, i.e.  $\pi \geq £10$ , and can never be worse than showing only one of the two. Similarly showing three means max at least second-<sup>fav</sup> best, so  $\pi \geq £15$ .

So exactly 3 is weakly best; <sup>weakly dominates</sup> in some sense it ~~shows~~ showing only 2 or 7 shoes, and is strictly better than showing 4.

g)

d) You need to exclude one shoe. So let's just compare  $E[\pi]$  in each case; given clients always choose their favourite of available ones.

$$\text{No A: } 0.5 \times (5+10) + 0.3 \times (5+15) + 0.2 \times (5+15)$$

good

Actually... since 3 shoes available, you either sell favourite or second-favourite. So we should exclude shoe which is nobody's favourite and can get first-best profits anyway.

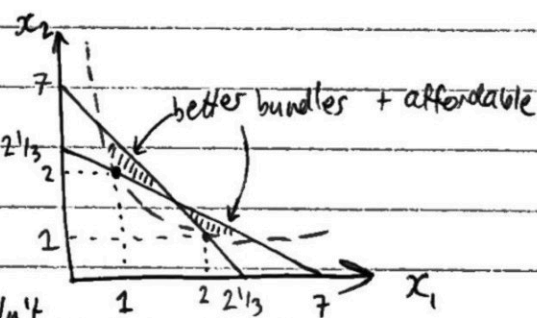
$\therefore$  exclude C; show  $\{A, B, D\}$  and always get £20 profit.

5. a)  $\nexists$  if  $c(3, 1) = (2, 1)$  then  $(2, 1) \succsim (1, 2)$  since the second bundle was affordable at that price vector. And similarly  $\nexists$  from  $\tilde{p}^*$  we get  $(1, 2) \succsim (2, 1)$ , i.e.  $(1, 2) \sim (2, 1)$ .

But this violates transitivity: monotonicity tells us  $(1, 3) \succsim (1, 2) \sim (2, 1)$  yet under the initial prices  $p = (3, 1)$  the bundle  $(1, 3)$  was affordable since cheaper than  $(2, 2)$  - but not purchased!

So WARP for bundles fails: if  $\underline{x}_1$  is chosen while  $\underline{x}_2$  affordable, then you cannot choose  $\underline{x}_2$  unless  $\underline{x}_1$  no longer affordable. But both

bundles affordable at both times, so shouldn't have switched.



For  $(2, 1)$  and  $(7/4, 7/4)$  we have  $p_1 x_2 = 7 \leq p_1 x_1 = 7$  so WARP requires that  $p_2 x_1 \geq p_2 x_2$

$(\Rightarrow) 5 \geq 7$  which is false; so again WARP violated and we couldn't see this because choice induced by rational preferences satisfies WARP.

- b)  $p_1 x_1 = 6$ ;  $p_1 x_2 = 5$  so  $x_2$  is affordable at  $(p_1, m_1)$   
 $p_2 x_1 = 10$ ;  $p_2 x_2 = 7$  but  $x_1$  not affordable at  $(p_2, m_2)$

$\therefore$  no violation of WARP; we could observe these.

- c) Strictly convex preferences mean ~~overlapping indifference curves~~ ~~indifference curves~~ ~~specifically~~ for some  $x \succ y$ , then  
 $x \succ \alpha x + (1-\alpha)y \succ y$  for all  $\alpha \in (0,1)$ .

The bundles in (b) are  $(1,3) \succ (1,2)$ . This is entirely consistent with strict convexity - along the line segment connecting them, all that changes is the quantity of good 2, and by monotonicity that means bundles closer to  $(1,2)$  are dispreferred, as required.

These couldn't be Cobb-Douglas preferences, however. For that, the consumer must optimally spend constant shares of their income on each good. But e.g. they spend  $3/7$  on  $x_1$  at  $t=1$  and  $3/6$  at  $t=2$ .

- 6.a) Notice that  $(9,2) \sim (4,0)$  while  $(9,2) \succ (4,1)$ .

But this violates stationarity, which requires

$$(x,t) \succ (y,s) \Leftrightarrow (x,t+r) \succ (y,s+r).$$

So by Fishburn-Rubinstein, we can't represent these preferences with a discounted utility function.

Suppose there was such a  $U$ ,  $U(x,t) = \delta^t u(x)$ . It would need to satisfy stationarity, since  $U(x,t+r) = \delta^r \cdot \delta^t u(x)$  and both sides of  $\succ$  are just multiplied by  $\delta^r$ , preserving  $\succ$ .

- b) We want a  $\delta$  s.t.

$$3 > \frac{3}{\delta+1} = 2 > \frac{3}{1/2+1} = \frac{3}{3/2} = 2 > \frac{2}{\delta+1} = \frac{2}{1/2+1} = \frac{2}{3/2} = \frac{4}{3} > \frac{2}{1/2+1} = \frac{2}{3/2} = \frac{4}{3} > \frac{1}{\delta+1} = \frac{1}{1/2+1} = \frac{2}{3} > \frac{1}{1/2+1} = \frac{2}{3} > \frac{1}{2}$$

pins down  $\delta = 1/2$

Checks it works for utilities prizes of all pairs of preferences

So  $\delta = 1/2$  works to capture the preferences perfectly.

The claim that discounted utility is "quasilinear in time" is a good intuition but slightly imprecise, quasilinearity usually refers to a specific functional form (linear in one argument). What you really mean is that the MRS under exponential discounting is independent of  $t$ .

This representation means that the agent is not "quasilinear" in time, as we allow the discount factor between periods to be a variable  $\frac{1+r_{t_1}}{1+r_{t_2}}$  rather than time-invariant ~~constant~~  $\delta^{(t_2-t_1)}$ .

✓ The model is of hyperbolic discounting, capturing that the agent cares most about the difference between "now" and "future", vs between two equivalently far-apart future times. Empirically, this has shown to be a more accurate model of how humans behave in time choice problems.

Good explainer